

Obstacle Detection using Ultrasonic Sensors

Prof. D. S. Vidhya

Assistant Professor (Guide)

*Department of Electronics and Telecommunication
Engineering
Don Bosco College of Engineering, Fatorda, India*

Miss Delicia Perlin Rebelo

UG Student

*Department of Electronics and Telecommunication
Engineering
Don Bosco College of Engineering, Fatorda, India*

Miss Cecilia Jane D'Silva

UG Student

*Department of Electronics and Telecommunication
Engineering
Don Bosco College of Engineering, Fatorda, India*

Mr. Linford William Fernandes

UG Student

*Department of Electronics and Telecommunication
Engineering
Don Bosco College of Engineering, Fatorda, India*

Miss Clarissa Joella Costa

UG Student

*Department of Electronics and Telecommunication Engineering
Don Bosco College of Engineering, Fatorda, India*

Abstract

In many applications like vehicle control, medical applications, robotic movement control, etc.; distance measurement of an object is used. This can be done using a variety of sensors- Ultrasonic, IR, radar, laser, etc. Measurement using ultrasonic sensors is the cheapest and its reliability among several others is very high. In this paper we discuss the uses of these sensors in distance measurement in vehicular applications and compare them for a small automobile prototype using Raspberry Pi, to provide an output for obstacle detection.

Keywords: Sensors; Ultrasonic Sensors; Distance Measurement; Obstacle Detection

I. INTRODUCTION

With the increasing demand for autonomous projects, the use of sensors is increasing. Sensors are sophisticated devices that convert the physical parameter (for example: temperature, pressure, humidity, speed, etc.) into a signal which can be measured electrically. They are very important to robotics. It is able to give robots remote access and make decisions as for a desired environment. It can also perceive its own environment and through programming can get the output it desires. Out of these, the motion detectors are based on the Infra-Red, Ultrasonic, and Microwave / radar technology. [1]

In industrial applications, ultrasonic sensors are characterized by their reliability and outstanding versatility. Ultrasonic sensors can be used to solve even the most complex tasks involving object detection or level measurement with millimeter precision, because their measuring method works reliably under almost all conditions. Infrared sensors too, find applications in many everyday products. Their low power requirements, their simple circuitry and their portable features make them desirable.[2]

II. CRITERIA TO CHOOSE A SENSOR

With the large number of sensors available in the market, it is necessary to choose the right sensor. There are certain features which have to be considered when we choose a sensor: Accuracy, Environmental condition, Range, Calibration, Resolution, Cost and Repeatability. [3]

In our project we have made a small prototype of an automobile for Adaptive Cruise Control where the car prototype, whose heart will be the Raspberry Pi, a microcontroller, which when set, will lock the object/car in front of it. For this we need to select a suitable sensor. Therefore, we first carried out a study of the following ultrasonic sensors.

A. Features of Ultrasonic Sensors

No other measuring method can be successfully put to use on such a wide scale and in so many different applications.

- 1) The device is extremely robust, making it suitable for even the toughest conditions.
- 2) The sensor surface cleans itself through vibration, making the sensor insensitive to dirt.
- 3) The physical principle—the propagation of sound—works, with a few exceptions, in practically any environment.
- 4) Ultrasonic sensors have proven their reliability and endurance in virtually all industrial sectors.

III. WORKING PRINCIPLE OF AN ULTRASONIC SENSOR

The ultrasonic sensor transmits sound waves and receives sound reflected from an object. When ultrasonic waves are incident on an object, diffused reflection of the energy takes place over a wide solid angle which might be as high as 180 degrees. Thus some fraction of the incident energy is reflected back to the transducer in the form of echoes. If the object is very close to the sensor, the sound waves returns quickly, but if the object is far away from the sensor, the sound waves takes longer to return. But if objects are too far away from the sensor, the signal takes so long to come back (or is very weak when it comes back) that the receiver cannot detect it.[4]

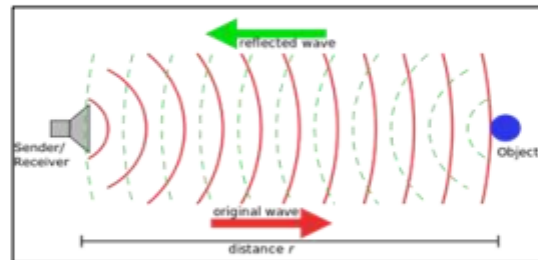


Fig. 1:

The sensor uses the time it takes for the sound to come back from the object in front to determine the distance of an object.

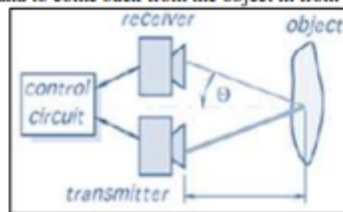


Fig. 2:

$$L = \frac{v t \cos \theta}{2}$$

The distance to the object (L) can then be calculated through the speed of ultrasonic waves (v) in the medium by the relation, where, 't' is the time taken by the wave to reach back to the sensor and 'θ' is the angle between the horizontal and the path taken as shown in the figure. If the object is in motion, instruments based on Doppler shift are used. The ultrasonic sensor can measure distances in centimetres and inches. It can measure from 0 to 2.5 meters, with a precision of 3 cm.

A. HCSR04 range sensor

HC-SR04 is a commonly used module for non-contact distance measurement for distances from 2cm to 400cm.

It has 4 pins:

VCC – 5V, input power; TRIG – Trigger Input; ECHO – Echo Output; GND – Ground



Fig. 3:

A trigger signal is provided to TRIG input, a HIGH signal of at least 10μs duration. This enables the module to transmit eight 40KHz ultrasonic burst. If there is an obstacle in-front of the module, it will reflect those ultrasonic waves. If the signal comes back, the ECHO output of the module will be HIGH for duration of time taken for sending and receiving ultrasonic signals. The

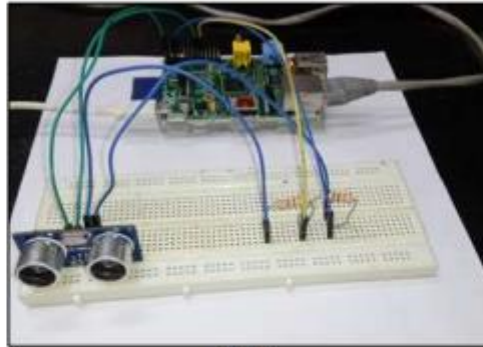


Fig. 6:

Suppose some obstacle is detected in front of the prototype, the ultrasonic sensor will detect a receiving signal and finds the approximate distance of the object from the prototype.

Table - 1

Sr. No.	Actual Distance (cm)	Measured Distance (cm)	Percentage error %
1	5	4.94	0.012
2	8	7.86	0.0175
3	10	9.98	0.002
4	13	13.19	0.015
5	15	14.82	0.012
6	18	18.2	0.011
7	20	20.03	0.0015
8	23	23.17	0.007
9	25	24.97	0.0012
10	28	27.87	0.0048

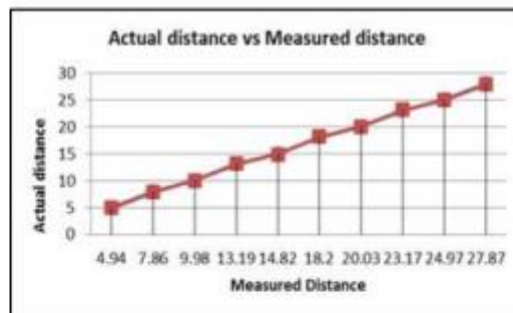


Fig. 7:

The experimental results for the distance measurement are shown in Table I. Fig. 7 shows the graph between actual distance and measured distance. We observe that there is considerable error in the measured distance as compared to the actual distance. The %error column also shows similar results. Since the error is very small, we can easily correct it while programming the code.

B. Obstacle Detection

On detection of the obstacle, the braking force to be applied depends upon the distance. It depends on the cases discussed below:

1) Case 1: The Obstacle is Stationary on the Path

In this case, the host vehicle detects the obstacle and determines the distance of the obstacle from the host. The speed is then, decreased automatically. If the distance exceeds the critical distance (not safe distance for driving); the braking mechanism is activated and the horn is pressed. If the obstacle is a living creature, it might move out of the path in accordance to the horn. But in case the obstacle is not moving, the speed is kept on decreasing in such a way that the host is brought to a stop at a fixed pre-set value before the obstacle.